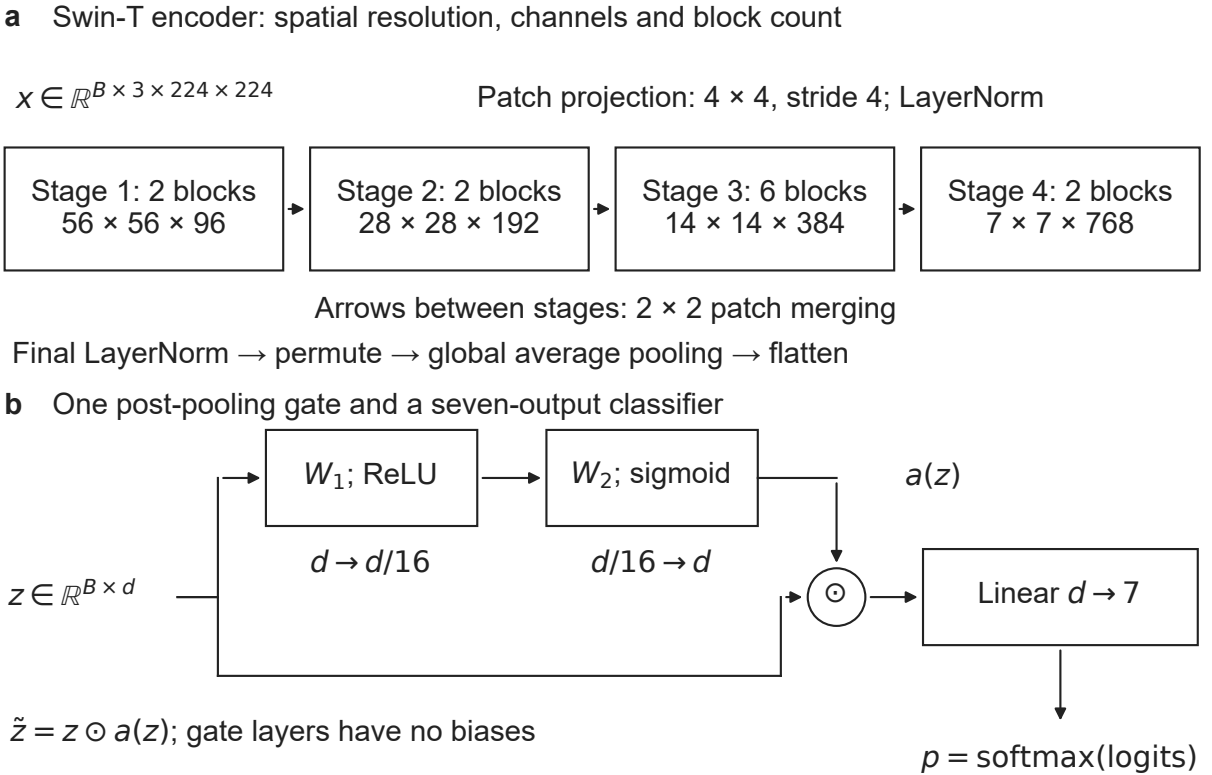

a Swin-T encoder: spatial resolution, channels and block count
$x \in \mathbb{R}^{B \times 3 \times 224 \times 224}$
Patch projection: 4 × 4, stride 4; LayerNorm
Stage 1: 2 blocks
56 × 56 × 96
Stage 2: 2 blocks
28 × 28 × 192
Stage 3: 6 blocks
14 × 14 × 384
Stage 4: 2 blocks
7 × 7 × 768
Arrows between stages: 2 × 2 patch merging
Final LayerNorm → permute → global average pooling → flatten
b One post-pooling gate and a seven-output classifier
W_1; ReLU
W_2; sigmoid
$a(z)$
$d \to d/16$
$d/16 \to d$
$z \in \mathbb{R}^{B \times d}$
Linear $d \to 7$
$\tilde{z} = z \odot a(z)$; gate layers have no biases
$p = \text{softmax}(\text{logits})$
Classifier includes a bias; no intermediate-stage gates
Swin-T: d = 768, hidden = 48 | ResNet-18: d = 512, hidden = 32